

ABSTRACT

With the rapid growth of artificial intelligence and human–computer interaction, conversational agents have evolved beyond text-based interfaces into immersive, multimodal systems. This project presents the design and implementation of a **3D virtual chatbot assistant** that integrates **machine learning–based natural language processing**, **speech synthesis**, and **real-time 3D animation** to provide a human-like interactive experience. The system accepts textual input from users, processes it using a large language model to generate contextual responses, converts the generated text into natural-sounding speech, and synchronizes the audio with a 3D humanoid avatar using lip-sync and facial expressions.

The backend is developed using Node.js and Express, incorporating an AI language model for response generation, a text-to-speech engine for audio synthesis, and an automated lip-sync generation pipeline. The frontend is implemented using React and Three.js, where a 3D avatar is rendered and animated in real time based on the chatbot's responses. Facial expressions, body animations, and viseme-based lip movements are dynamically applied to enhance realism.

The proposed system demonstrates an effective combination of artificial intelligence, web technologies, and computer graphics to create an engaging virtual assistant. The results show smooth conversational flow, synchronized audio-visual output, and an interactive user interface, making the system suitable for applications such as virtual assistants, digital companions, interactive learning environments, and entertainment systems.

CHAPTER 1

INTRODUCTION

Conversational agents, commonly referred to as chatbots, have become an integral part of modern digital systems. Traditional chatbots primarily rely on text-based interaction, which limits user engagement and emotional expressiveness. Recent advancements in artificial intelligence, speech synthesis, and 3D graphics have enabled the development of more immersive virtual assistants capable of mimicking human-like behavior.

This project focuses on the development of a **3D virtual chatbot assistant** that combines machine learning–driven conversation with speech and realistic avatar animation. Instead of responding only with text, the system delivers spoken responses accompanied by facial expressions, lip movements, and body animations rendered on a 3D humanoid model. This significantly enhances the sense of presence and realism during interaction.

1.1 Objectives of the Project

- To design an AI-powered chatbot capable of generating contextual responses.
- To convert textual responses into natural-sounding speech.
- To synchronize speech with accurate lip movements using phoneme-based lip-sync.
- To animate a 3D humanoid avatar with facial expressions and gestures.
- To build a responsive web-based interface for user interaction.

1.2 Scope

The scope of this project includes conversational AI, speech synthesis, lip-sync generation, and real-time 3D rendering in a browser environment. The system is intended for educational and demonstrative purposes and can be extended for real-world virtual assistant applications.

CHAPTER 2

LITERATURE REVIEW

Conversational agents have evolved significantly from early rule-based systems to modern data-driven models powered by deep learning. Traditional chatbots relied on predefined rules and templates, which limited their ability to handle open-domain conversations and contextual variations. The emergence of large-scale neural language models marked a turning point in conversational AI research.

Brown *et al.* demonstrated that large language models trained on massive datasets are capable of performing a wide range of natural language tasks, including dialogue generation, using few-shot learning techniques [1]. This work laid the foundation for modern conversational systems by showing that a single model can generalize across tasks with minimal fine-tuning. Building upon this, Adiwardana *et al.* focused on improving the human-likeness of open-domain chatbots, emphasizing response quality, safety, and conversational coherence [2].

Although conversational intelligence has improved, many systems remain limited to text-based interaction. Research on embodied conversational agents highlighted the importance of visual and non-verbal communication in human–computer interaction. Cassell *et al.* introduced the concept of embodied conversational agents, where speech, facial expressions, and gestures are combined to create more natural and engaging interactions [3]. Further studies by Kopp *et al.* reinforced that socially situated virtual agents with expressive behaviors significantly enhance user engagement and perceived intelligence [4].

Speech synthesis is another critical component of interactive virtual assistants. Early speech synthesis techniques produced robotic and unnatural voices. Shen *et al.* proposed a neural text-to-speech architecture that generates high-quality, natural-sounding speech by conditioning waveform generation on spectrogram predictions [5]. More recently, Tan *et al.* presented a comprehensive survey of neural speech synthesis systems, highlighting advancements in expressiveness, prosody control, and naturalness achieved by modern deep learning approaches [6].

Facial animation and lip synchronization play a vital role in creating believable virtual humans. Taylor *et al.* explored audio-driven facial animation techniques that enable realistic lip movements directly from speech signals, reducing the need for manual animation [7]. Suwajanakorn *et al.* demonstrated high-fidelity lip synchronization by learning mappings from audio to facial motion, proving that accurate lip-sync can be achieved using speech alone [8]. These works provide a strong foundation for integrating lip-sync mechanisms into real-time systems.

From a human–computer interaction perspective, the use of 3D avatars has been shown to positively influence user engagement and trust. Nowak and Fox studied the impact of avatars on online interaction and found that visual embodiment enhances social presence and user

involvement [9]. Similarly, Bailenson *et al.* demonstrated that virtual humans improve learning outcomes and user immersion in interactive systems [10].

Web-based 3D rendering technologies have enabled the deployment of interactive 3D applications directly in browsers. Evans *et al.* discussed the role of WebGL in enabling hardware-accelerated 3D graphics on the web, making real-time 3D applications more accessible [11]. This advancement allows modern virtual assistants to combine AI, speech, and 3D animation within a single web-based platform.

Despite extensive research in conversational AI, speech synthesis, facial animation, and virtual avatars, limited work has focused on integrating all these components into a unified, real-time, web-based system. This project addresses this research gap by combining machine learning-based conversation, neural speech synthesis, lip synchronization, and real-time 3D avatar animation to create an immersive virtual chatbot assistant.

CHAPTER 3

METHODOLOGY

3.1 System Architecture

The system follows a client–server architecture consisting of:

- **Backend Server:** Handles AI response generation, speech synthesis, and lip-sync data creation.
- **Frontend Client:** Manages user interaction, 3D rendering, audio playback, and animation synchronization.

3.2 Backend Implementation

The backend is developed using Node.js and Express. It exposes a REST API endpoint that accepts user messages and returns structured chatbot responses.

- **Conversational AI**
User input is processed using an AI language model provided by **OpenRouter**, which generates context-aware responses based on conversational history and predefined system instructions.
- **Speech Synthesis**
The generated text is converted into speech using **ElevenLabs** text-to-speech technology, producing high-quality audio output.
- **Lip-Sync Generation**
The synthesized audio is converted into a waveform format and processed using a lip-sync tool that extracts phoneme timing information. This information is stored as JSON data containing time-stamped mouth cues.

Backend Output

Each chatbot response includes:

- Textual message
- Base64-encoded audio
- Lip-sync JSON data
- Facial expression identifier
- Animation name

3.3 Frontend Implementation

The frontend is built using React and **Three.js** with React Three Fiber for declarative 3D rendering.

3D Avatar Rendering

A humanoid avatar model stored in GLB format is loaded and rendered in real time. The model includes a skeleton, morph targets, and materials required for facial animation.

Facial Expressions and Animations

Facial expressions are implemented using morph target interpolation, allowing smooth transitions between emotions such as happiness, sadness, anger, and surprise. Body animations such as idle, talking, and laughing are applied using animation mixers.

Lip Synchronization

Lip-sync is achieved by mapping phoneme cues from the backend to corresponding visemes on the avatar's mouth. During audio playback, morph targets are updated frame-by-frame to match speech timing.

User Interface

The UI allows users to enter text, send messages, control camera zoom, and toggle visual modes. State management is handled using React Context and custom hooks to ensure synchronized audio and animation playback.

CHAPTER 4

RESULTS AND CONCLUSION

4.1 Results

The implemented system successfully demonstrates real-time interaction between the user and the virtual assistant. The chatbot generates coherent and contextually appropriate responses, which are converted into expressive spoken output. Lip synchronization between audio and avatar mouth movements is smooth and visually convincing.

Facial expressions and animations enhance emotional realism, making interactions more engaging compared to traditional text-based chatbots. The web-based deployment ensures accessibility without requiring specialized hardware or software.

Performance testing shows acceptable response times, with most delays attributed to AI response generation and audio synthesis. Overall, the system achieves its intended objectives effectively.

4.2 Conclusion

This project successfully integrates machine learning, speech synthesis, and 3D animation to create an interactive **3D virtual chatbot assistant**. The system demonstrates how conversational AI can be enhanced through visual embodiment and audio-visual synchronization.

The project provided practical exposure to backend API development, AI integration, real-time 3D rendering, and state management in modern web applications. While the current implementation is suitable for demonstration and educational use, it can be further enhanced with emotion detection, multilingual support, and improved animation realism.

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